

VIVA QUESTION BOOKLET

Comprehensive VSA Study Guide

For Mini Project:

OBJECT FOLLOWING ROBOT USING ARDUINO UNO

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Topic 0: Basic Prerequisite Understanding & Knowledge

Q1. What is an embedded system?

Answer: An embedded system is a dedicated computer system built into a larger product to perform a specific function, such as reading sensors and controlling actuators.

Q2. What does the term “autonomous robot” mean?

Answer: An autonomous robot operates independently without direct human input, using sensors to detect its environment and make decisions on its own.

Q3. What does VCC stand for and what is its role?

Answer: VCC stands for Voltage at the Common Collector and represents the positive supply voltage rail that powers active components on the circuit board.

Q4. What does GND represent in an electronic circuit?

Answer: GND represents the electrical common ground or reference point at 0V, providing a shared return path for current from all powered components.

Q5. What does GPIO stand for and what is it used for?

Answer: GPIO stands for General Purpose Input/Output; these configurable pins on a microcontroller can read digital sensor signals or drive output devices such as LEDs and motors.

Q6. What does PWM stand for, and what is it used to control?

Answer: PWM stands for Pulse Width Modulation; it varies the duty cycle of a digital square wave to control the average power delivered to a device, such as a DC motor.

Q7. What is a duty cycle in PWM?

Answer: The duty cycle is the ratio of the ON time of a pulse to its total period, expressed as a percentage; a higher duty cycle delivers more average power to the load.

Q8. What is the difference between an analog and a digital signal?

Answer: An analog signal is continuous and can take any value over a range, whereas a digital signal has only two discrete states: HIGH (1) or LOW (0).

Q9. What is a microcontroller, and how does it differ from a microprocessor?

Answer: A microcontroller integrates a CPU, memory, and I/O peripherals on a single chip for embedded applications, whereas a microprocessor is only a CPU that requires external memory and peripherals.

Q10. State Ohm's Law and write its mathematical expression.

Answer: Ohm's Law states that the current through a conductor is proportional to the voltage across it: $V = I \times R$, where V is voltage, I is current, and R is resistance.

Q11. What is an H-Bridge and what is its primary function?

Answer: An H-Bridge is a circuit of four switches that can reverse the polarity of voltage across a load, allowing a motor to spin in both forward and reverse directions.

Q12. What is an ultrasonic wave?

Answer: An ultrasonic wave is a sound wave with a frequency above 20 kHz, which is higher than the upper limit of human hearing.

Q13. What is infrared (IR) radiation?

Answer: Infrared radiation is electromagnetic energy with wavelengths longer than visible light (approximately 700 nm to 1 mm), used in remote sensing and proximity detection.

Q14. What is a gear reduction ratio, and why is it used in motors?

Answer: A gear reduction ratio decreases the output speed while proportionally increasing output torque; it allows small motors to drive heavier loads that require more turning force.

Q15. What does the term "differential drive" mean in robotics?

Answer: A differential drive is a steering system where two independent wheel pairs are driven at different speeds or directions to produce turning and linear motion.

Q16. What is a state machine in software design?

Answer: A state machine is a programming model that defines a set of conditions (states) and the transitions between them based on input events, enabling organized decision-making.

Q17. What is a short circuit, and why is it dangerous?

Answer: A short circuit is a low-resistance path between supply and ground that causes uncontrolled high current flow, leading to overheating, battery damage, or component destruction.

Q18. What is back-EMF in DC motors?

Answer: Back-EMF (Electromotive Force) is the voltage generated by the spinning motor acting like a generator; it opposes the supply voltage and can cause transient voltage spikes in the driver circuit.

Q19. What does the acronym “RISC” stand for?

Answer: RISC stands for Reduced Instruction Set Computing, an architecture design that uses a small set of simple instructions to increase processing speed and efficiency.

Q20. State Kirchhoff’s Voltage Law (KVL) and Current Law (KCL).

Answer: KVL states that the algebraic sum of all voltages around a closed loop is zero; KCL states that the sum of currents entering any node equals the sum leaving it.

Topic 1: Arduino UNO & Microcontroller Architecture

Q21. What microcontroller chip is used on the Arduino UNO?

Answer: The Arduino UNO is based on the ATmega328P, an 8-bit RISC microcontroller manufactured by Microchip Technology (formerly Atmel).

Q22. What is the clock frequency of the ATmega328P on the Arduino UNO?

Answer: The ATmega328P on the Arduino UNO runs at a clock frequency of 16 MHz using an external crystal oscillator.

Q23. How much Flash, SRAM, and EEPROM memory does the ATmega328P have?

Answer: The ATmega328P contains 32 KB of Flash memory (program storage), 2 KB of SRAM (runtime data), and 1 KB of EEPROM (non-volatile data storage).

Q24. How many digital I/O pins does the Arduino UNO have, and how many support PWM?

Answer: The Arduino UNO has 14 digital I/O pins, of which 6 are capable of generating hardware PWM output signals.

Q25. How many analog input pins does the Arduino UNO have?

Answer: The Arduino UNO has 6 analog input pins (A0 to A5), each connected to an onboard 10-bit Analog-to-Digital Converter (ADC).

Q26. What is the operating voltage of the Arduino UNO?

Answer: The Arduino UNO operates at 5V logic levels; it can be powered from 7V to 12V via the barrel jack or 5V via the USB port.

Q27. How do you configure a GPIO pin as a digital output in Arduino code?

Answer: You use the `pinMode(pin, OUTPUT)` function in the `setup()` block to configure the specified pin to drive digital output signals.

Q28. What is the purpose of the `setup()` and `loop()` functions in Arduino?

Answer: The `setup()` function runs once at startup to initialize pins and peripherals; the `loop()` function runs repeatedly in an infinite cycle to execute the main control logic.

Q29. What is the `analogWrite()` function and what range of values does it accept?

Answer: `analogWrite()` outputs a PWM signal on a PWM-capable pin; it accepts values from 0 (always LOW) to 255 (always HIGH), representing 0% to 100% duty cycle.

Q30. What is the `digitalRead()` function used for in this project?

Answer: It reads the current logic state (HIGH or LOW) of a digital input pin; in this project it reads the binary output signals from the left and right IR sensors.

Q31. What does the `delay()` function do in Arduino code?

Answer: The `delay(ms)` function pauses execution of the code for the specified number of milliseconds, blocking all other processing during that time.

Q32. What is the difference between `INPUT` and `INPUT_PULLUP` in Arduino?

Answer: `INPUT` configures a pin with no pull resistor, making it float when unconnected; `INPUT_PULLUP` enables an internal pull-up resistor, holding the pin HIGH when no external signal is applied.

Q33. What is the purpose of the `return` statement in the safety recovery block of the code?

Answer: The `return` statement exits the current iteration of the `loop()` function immediately after the safety backup maneuver, preventing any further sensor processing in that cycle.

Q34. What is the ADC resolution of the ATmega328P and what is its maximum integer output?

Answer: The ATmega328P has a 10-bit ADC, producing integer values from 0 to 1023 that map the analog input voltage range.

Q35. What is EEPROM, and what advantage does it offer in embedded systems?

Answer: EEPROM (Electrically Erasable Programmable Read-Only Memory) is non-volatile memory that retains data when power is removed, useful for storing calibration values permanently.

Q36. What is the difference between blocking code and non-blocking code?

Answer: Blocking code (such as using `delay()`) halts program execution until it completes; non-blocking code allows the loop to continue executing other tasks while waiting.

Q37. What is the Arduino `Serial.begin()` function used for?

Answer: It initializes the hardware UART serial port at a specified baud rate, enabling communication between the Arduino and a PC for debugging via the Serial Monitor.

Q38. What is the difference between `digitalWrite()` and `analogWrite()`?

Answer: `digitalWrite()` sets a pin fully HIGH or LOW; `analogWrite()` outputs a variable duty cycle PWM wave to simulate intermediate average voltage levels.

Q39. What is a watchdog timer and does the ATmega328P include one?

Answer: A watchdog timer automatically resets the microcontroller if software stops responding; yes, the ATmega328P includes a built-in watchdog timer with configurable timeout periods.

Q40. Why was the Arduino UNO chosen over an ESP32 for this project?

Answer: The Arduino UNO is simpler, lower in cost, and entirely sufficient for processing ultrasonic and IR sensor data; the project does not require wireless communication or high-speed computing.

Topic 2: HC-SR04 Ultrasonic Sensor

Q41. What is the operating voltage of the HC-SR04 sensor?

Answer: The HC-SR04 ultrasonic distance sensor operates at 5V DC, which is directly compatible with the Arduino UNO's output voltage.

Q42. What is the frequency of the ultrasonic burst emitted by the HC-SR04?

Answer: The HC-SR04 transmits an 8-cycle burst of ultrasonic waves at 40 kHz, which is well above the range of human hearing.

Q43. Write the distance formula used by the HC-SR04 and explain each variable.

Answer: $d = (t \times 0.034) / 2$, where t is the echo pulse duration in microseconds, $0.034 \text{ cm}/\mu\text{s}$ is the speed of sound, and the factor of 2 accounts for the round trip of the wave.

Q44. What is the effective measuring range of the HC-SR04?

Answer: The HC-SR04 can measure distances accurately from 2 cm to 400 cm (4 metres) with a stated accuracy of approximately 3 mm.

Q45. What pins of the HC-SR04 are connected to the Arduino UNO in this project?

Answer: The TRIG pin is connected to digital pin D8 and the ECHO pin is connected to digital pin D13 of the Arduino UNO.

Q46. How long is the trigger pulse that must be sent to the HC-SR04?

Answer: A $10 \mu\text{s}$ (microsecond) HIGH pulse must be sent on the TRIG pin to initiate a single ranging measurement cycle.

Q47. What does the NewPing library do, and why is it used in this project?

Answer: The NewPing library provides an optimized, non-blocking interface for the HC-SR04 sensor, handling trigger-echo timing internally and returning the measured distance directly in centimeters.

Q48. What does the code `sonar.ping_cm()` return?

Answer: It returns the measured distance in centimeters as an integer; it returns 0 if no echo is received within the maximum detection timeout period.

Q49. Why does the code replace a zero return value with 200 cm?

Answer: A zero return means no target was detected within range; replacing it with 200 cm ensures the robot enters search mode instead of triggering an incorrect safety recovery maneuver.

Q50. What is the speed of sound in air at room temperature?

Answer: The speed of sound in air at approximately 20°C is about 343 m/s (or 0.0343 cm/ μ s), which is the physical constant used in the distance formula.

Q51. Why is the HC-SR04 not affected by ambient lighting conditions?

Answer: The HC-SR04 uses acoustic (sound) waves rather than light, so changes in room brightness or direct sunlight do not interfere with its distance measurements.

Q52. What is the sensing beam angle of the HC-SR04?

Answer: The HC-SR04 has an effective cone-shaped sensing beam with an angle of approximately 15 degrees around its central axis.

Q53. What is the minimum measurable distance of the HC-SR04, and why does it have a lower limit?

Answer: The minimum distance is 2 cm; objects closer than this fall within the transition zone where the transmitted pulse overlaps with the echo before the sensor can switch modes.

Q54. What is the purpose of defining `MAX_DISTANCE 200` in the code?

Answer: It sets the NewPing library's timeout limit, instructing the library to stop listening for echoes after the time equivalent to a 200 cm round trip has elapsed.

Q55. What is the control loop latency achieved in this project, and why does it matter?

Answer: The control loop latency was measured at approximately 10 ms, enabling the robot to react to target movements in near-real time without noticeable lag.

Q56. What is the difference between the TRIG and ECHO pins on the HC-SR04?

Answer: The TRIG pin is an input that receives the 10 μ s trigger pulse from the Arduino; the ECHO pin is an output that produces a high pulse whose duration encodes the measured distance.

Q57. Why should the HC-SR04 not be tested with soft or fabric surfaces?

Answer: Soft materials absorb ultrasonic energy and produce weak echoes, significantly reducing detection accuracy and potentially causing missed or erroneous distance readings.

Q58. Can two HC-SR04 sensors interfere with each other on the same robot?

Answer: Yes, if both sensors transmit simultaneously, one sensor's echo signal can be intercepted by the other; they must be triggered sequentially to avoid cross-talk.

Q59. What does the ultrasonic sensor represent in the system block diagram?

Answer: It forms the primary sensor input, feeding Echo/Trig signals into the Arduino UNO to determine the forward distance to the target.

Q60. How does the code determine whether to apply a high or low speed PWM value?

Answer: The ternary expression (distance > 25) ? 220 : 170 sets PWM to 220 when the target is more than 25 cm away and to 170 when it is within 25 cm, slowing the robot as it approaches.

Topic 3: IR Proximity Sensors & Lateral Tracking

Q61. How does an IR proximity sensor detect an obstacle?

Answer: The sensor emits infrared light from an IR LED; when an object is nearby, reflected light reaches the photodiode receiver. An LM393 comparator outputs a digital LOW when reflected signal strength exceeds the set threshold.

Q62. What is the output logic of the IR sensor when an object is detected?

Answer: The IR sensor outputs a digital LOW (logic 0) when an obstacle is detected within its calibrated range, and a HIGH (logic 1) when no obstacle is present.

Q63. How is the detection range of the IR sensor adjusted?

Answer: The onboard potentiometer adjusts the reference voltage of the LM393 comparator, setting the sensitivity and maximum detection distance (calibrated to approximately 8 cm in this project).

Q64. What is the typical detection range calibrated for the IR sensors in this project?

Answer: The IR sensors were calibrated to trigger when an object was within approximately 8 cm, preventing false detections from ambient light and distant surfaces.

Q65. Which Arduino UNO pins are the right and left IR sensor outputs connected to?

Answer: The Right IR sensor output is connected to D2 and the Left IR sensor output is connected to D3 of the Arduino UNO.

Q66. What action does the robot take when only the right IR sensor detects an object?

Answer: When the right IR sensor outputs LOW (`rightIR == LOW`) and the left is HIGH, the robot executes `setMotors('R')` to pivot right toward the target.

Q67. What action does the robot take when only the left IR sensor detects an object?

Answer: When the left IR sensor outputs LOW (`leftIR == LOW`) and the right is HIGH, the robot executes `setMotors('L')` to pivot left toward the target.

Q68. What happens when both IR sensors read HIGH simultaneously?

Answer: Both HIGH indicates no lateral obstacle; the robot then uses the ultrasonic distance to decide whether to drive forward (10–35 cm) or enter search mode (more than 35 cm).

Q69. Why are IR sensors used for lateral tracking instead of a second ultrasonic sensor?

Answer: IR sensors are smaller, simpler, and cheaper; their binary detection output is sufficient for lateral deviation detection, and adding a second ultrasonic sensor would require complex angular scanning logic.

Q70. Why are IR sensors susceptible to ambient sunlight interference?

Answer: Direct sunlight contains a strong infrared component that can saturate the photodiode receiver, preventing it from distinguishing between sunlight and the sensor's own reflected IR signal.

Q71. What is the LM393 comparator in the IR sensor module?

Answer: The LM393 is a dual voltage comparator IC that compares the photodiode output against a reference set by the potentiometer and generates a clean digital HIGH or LOW signal.

Q72. What is the purpose of the indicator LED on the IR sensor module?

Answer: The indicator LED lights up when an obstacle is detected, providing a visual diagnostic tool to verify sensor triggering during calibration without needing a serial monitor.

Q73. If the left IR sensor output reads LOW, which side is the target on?

Answer: A LOW reading on the left IR sensor means the target has been detected on the left flank of the robot, prompting it to steer left.

Q74. What is the condition in the code for the robot to drive forward?

Answer: The robot drives forward when `distance >= MIN_DISTANCE && distance <= FOLLOW_DISTANCE` is true and neither IR sensor is detecting a lateral object.

Q75. What does the code condition `rightIR == HIGH && leftIR == LOW` trigger?

Answer: This condition means the left IR sensor has detected the target, so the robot calls `setMotors('L')` to pivot left to follow it.

Q76. What is the role of IR sensors in the system architecture block diagram?

Answer: They form the lateral sensor input layer, feeding binary Digital In signals to the Arduino UNO that indicate whether the target is on the left or right flank.

Q77. Can the IR sensor be used in complete darkness?

Answer: Yes, IR sensors are self-contained; they emit their own infrared light and measure reflections, making them fully functional in complete darkness.

Q78. What is the effect of a highly reflective or shiny surface on IR sensor performance?

Answer: Highly reflective surfaces return a stronger echo, potentially triggering detection at a greater distance than the calibrated range and causing the robot to respond incorrectly.

Q79. Why is it important to calibrate both IR sensors symmetrically?

Answer: Asymmetric calibration would cause one sensor to trigger at a greater range than the other, making the robot consistently bias its steering to one side even when following a straight-ahead target.

Q80. What is the search mode behavior of the robot when no sensor detects a target?

Answer: When the ultrasonic distance exceeds 35 cm and both IR sensors read HIGH, the `else` branch calls `setMotors('R')`, making the robot pivot right to scan the environment for the lost target.

Topic 4: L298N Motor Driver & Power Scheme

Q81. What does the L298N contain internally to drive motors?

Answer: The L298N contains two full H-bridge circuits built from high-power bipolar transistors (BJTs), allowing it to independently control two DC motors in both directions.

Q82. What are the ENA and ENB pins on the L298N, and how are they used?

Answer: ENA and ENB are the enable pins for the two H-bridge channels; applying a PWM signal to them regulates the average motor voltage and controls motor speed.

Q83. What are the IN1, IN2, IN3, and IN4 pins on the L298N?

Answer: These are direction control input pins; setting IN1 HIGH and IN2 LOW spins the first motor forward, while IN1 LOW and IN2 HIGH reverses it. IN3/IN4 control the second motor similarly.

Q84. What is the maximum supply voltage and current capacity of the L298N?

Answer: The L298N can handle motor supply voltages up to 46V and deliver a peak output current of 2A per channel.

Q85. Why is the 7.4V battery pack connected directly to the L298N and not directly to the Arduino?

Answer: To prevent motor current spikes from causing voltage drops on the microcontroller's supply rail, which would cause the Arduino to reset unexpectedly.

Q86. How does the Arduino UNO receive its 5V supply in this project?

Answer: The onboard LM7805 voltage regulator on the L298N driver board steps down the 7.4V motor supply to a stable 5V output, which then powers the Arduino UNO.

Q87. What is the function of connecting all grounds together in the circuit?

Answer: Connecting the grounds of the battery, Arduino, and L298N establishes a shared 0V reference point, ensuring that PWM control signals from the Arduino are correctly interpreted by the driver.

Q88. What problem was observed when the SG90 servo motor was included in the prototype?

Answer: The SG90 servo motor drew high inrush current during rotation, causing a voltage brownout on the 5V logic rail that repeatedly reset the Arduino and destabilized sensor readings.

Q89. Why was the OLED display removed from the final prototype?

Answer: The OLED display was removed as part of the power simplification; it added to the logic rail current load and, combined with the servo, contributed to voltage instability causing repeated resets.

Q90. What is a voltage brownout in microcontroller circuits?

Answer: A brownout occurs when the supply voltage drops below the minimum operating threshold of the microcontroller, causing it to reset, malfunction, or produce erratic sensor readings.

Q91. How are the four BO motors wired in this project?

Answer: The four motors are wired as two parallel pairs: the left two motors connected to channel A (ENA, IN1, IN2) and the right two motors connected to channel B (ENB, IN3, IN4).

Q92. What is the consequence of not connecting the common ground between the Arduino and L298N?

Answer: Without a common ground, the PWM signals from the Arduino have no shared voltage reference with the driver, causing erratic motor behavior or complete control failure.

Q93. What is back-EMF suppression in the L298N, and how is it handled?

Answer: When motors stop or reverse, they generate inductive back-EMF spikes; the L298N includes built-in freewheeling diodes that provide a safe return path for this energy, protecting the transistors.

Q94. How does the PWM value of 170 vs. 220 affect the robot's movement speed?

Answer: A PWM value of 170 ($\approx 67\%$ duty cycle) delivers less average voltage to the motors for slower speed when close to the target; 220 ($\approx 86\%$) delivers higher speed when the target is farther away.

Q95. What is the difference between motor “stop” and motor “brake” in H-Bridge control?

Answer: Stop (coasting) removes drive current, letting the motor slow under friction; braking shorts the motor terminals to create regenerative braking that stops it quickly.

Q96. What is the LM7805 IC, and where is it used in this circuit?

Answer: The LM7805 is a linear voltage regulator outputting a stable 5V from a higher input; it is integrated on the L298N driver board to power the Arduino UNO's logic.

Q97. What future upgrade is proposed to fix the servo motor voltage problem?

Answer: Adding an LM2596 step-down (buck) converter to create an isolated, regulated power rail for the servo, preventing its current spikes from affecting the Arduino's 5V logic rail.

Q98. What is the operating voltage of the BO geared motors used in this project?

Answer: The BO geared motors operate between 3V and 9V DC, driven at approximately 6V (after the L298N internal voltage drop) from the 7.4V battery supply.

Q99. What is the RPM and gear reduction ratio of the BO motors used?

Answer: The motors deliver approximately 150 RPM at 6V and have a 1:48 gear reduction ratio, trading rotational speed for increased torque to drive the chassis.

Q100. What is the approximate battery runtime achieved in this project?

Answer: The dual 18650 Lithium-Ion cell pack (7.4V, 2200 mAh) supplied stable power for approximately 50 minutes of active robot operation under full motor load.

Topic 5: Differential Drive Kinematics & Control Algorithm

Q101. What is differential drive kinematics?

Answer: Differential drive kinematics describes how a robot's linear and angular velocities result from varying the speeds of its independently controlled left and right wheel pairs.

Q102. Write the equation for the linear velocity of a differential drive robot.

Answer: The linear velocity is $v = \frac{v_R + v_L}{2}$, where v_R and v_L are the linear speeds of the right and left wheels respectively.

Q103. Write the equation for the angular velocity of a differential drive robot.

Answer: The angular velocity is $\omega = \frac{v_R - v_L}{W}$, where W is the wheel track width (distance between the left and right wheels).

Q104. How does the robot execute a forward motion?

Answer: Both the left and right wheel pairs rotate forward at equal speeds ($v_L = v_R > 0$), producing zero angular velocity and purely linear forward motion.

Q105. How does the robot execute a backward (reverse) motion?

Answer: Both the left and right wheel pairs rotate in reverse at equal speeds ($v_L = v_R < 0$), producing backward linear motion with no rotation.

Q106. How does the robot execute a pivot left turn?

Answer: The left wheels rotate in reverse while the right wheels rotate forward ($v_R = -v_L > 0$), creating equal and opposite forces that rotate the robot in place to the left.

Q107. How does the robot execute a pivot right turn?

Answer: The right wheels rotate in reverse while the left wheels rotate forward ($v_L = -v_R > 0$), creating equal and opposite forces that rotate the robot in place to the right.

Q108. What direction code does the character ‘F’ represent in the `setMotors()` function?

Answer: The character ‘F’ sets IN1 and IN3 HIGH (and IN2, IN4 LOW), driving both motor pairs in the forward direction.

Q109. What direction code does the character ‘B’ represent in the `setMotors()` function?

Answer: The character ‘B’ sets IN2 and IN4 HIGH (and IN1, IN3 LOW), reversing both motor pairs to drive the robot backward.

Q110. What direction code does the character ‘S’ represent, and how is it implemented?

Answer: The character ‘S’ represents Stop; all four IN pins default to LOW (the default: case does nothing), delivering zero drive current to both motor pairs.

Q111. What does the safety recovery sequence do, step by step?

Answer: When distance < 10 cm: stop (S) for 300 ms, reverse (B) for 400 ms, stop (S) for 200 ms, pivot right (R) for 300 ms, then stop (S) and exit the loop with return.

Q112. What is the `MIN_DISTANCE` threshold defined in the code and what does it trigger?

Answer: `MIN_DISTANCE` is defined as 10 cm; if the target comes closer than this, it triggers the safety backup and turn maneuver to prevent a collision.

Q113. What is the `FOLLOW_DISTANCE` threshold and what behavior does it bound?

Answer: `FOLLOW_DISTANCE` is defined as 35 cm; the robot drives forward when the target distance is between 10 cm and 35 cm. Beyond 35 cm the robot enters search mode.

Q114. What is dynamic speed control and how is it implemented in the code?

Answer: Dynamic speed control adjusts motor PWM based on distance; the ternary expression `(distance > 25) ? 220 : 170` applies a higher duty cycle when the target is farther away.

Q115. Why is a C++ `switch` statement used in `setMotors()` instead of an `if-else` chain?

Answer: A switch statement produces cleaner, more readable code for multiple discrete cases and may compile to slightly more efficient jump-table instructions on the ATmega328P.

Q116. What role does the NewPing object play at the top of the code?

Answer: `NewPing sonar(TRIG_PIN, ECHO_PIN, MAX_DISTANCE)` creates a sonar object that internally manages the TRIG pulse timing and ECHO measurement for the HC-SR04.

Q117. What five sensor state conditions are mapped in the calibration table?

Answer: (1) Distance <10 cm: Safety Back & Turn; (2) 10–35 cm, both IR HIGH: Forward; (3) Right IR LOW: Pivot Right; (4) Left IR LOW: Pivot Left; (5) >35 cm, both IR HIGH: Search Pivot.

Q118. What ranging accuracy was measured during testing?

Answer: The ranging accuracy was ± 0.5 cm within the active tracking envelope of 10 cm to 35 cm during the indoor prototype evaluation.

Q119. Why does the safety distance check occur before the IR sensor checks in the code?

Answer: A collision risk is safety-critical and must override all other behaviors; placing it first ensures the robot always backs off before any steering decision is made.

Q120. What is the key advantage of using a state machine in the robot's control code?

Answer: A state machine provides clear, organized, and mutually exclusive behavior modes (Forward, Left, Right, Backup, Search), making the code easy to read, debug, and extend.

Topic 6: System Integration & Experimental Results

Q121. What is shown in the system block diagram of this robot?

Answer: The block diagram shows the HC-SR04 and dual IR sensors feeding signals to the Arduino UNO, which outputs PWM and direction signals to the L298N driver, which drives the four BO motors; the 7.4V battery powers both the driver and the Arduino.

Q122. What software framework was used to write the control firmware?

Answer: The control firmware was written in C++ using the Arduino IDE 2.x framework, leveraging the `Arduino.h` core library and the `NewPing.h` sensor library.

Q123. Why was the NewPing library selected over a manual `pulseIn` implementation?

Answer: NewPing uses non-blocking timing and handles edge cases such as no-echo returns, providing a cleaner and more reliable abstraction than writing raw `pulseIn()` and `delayMicroseconds()` calls.

Q124. How was the IR sensor calibration performed during system testing?

Answer: The onboard potentiometers were adjusted while placing an object at exactly 8 cm until the indicator LED reliably lit up, ensuring consistent detection across both sensors.

Q125. What chassis type does the robot use, and how many wheels does it have?

Answer: The robot uses a 4-wheel differential drive chassis with four BO geared motors wired in parallel left and right pairs for independent side control.

Q126. What is the battery configuration used in this project?

Answer: Two 18650 Lithium-Ion cells are connected in series, producing a 7.4V nominal output with a 2200 mAh capacity.

Q127. What performance envelope was validated during testing?

Answer: The prototype reliably followed targets between 10 cm and 35 cm, backed up to recover a safe distance if the target was closer than 10 cm, and steered smoothly for lateral movements.

Q128. What is the primary conclusion of this mini project?

Answer: The project demonstrates that a low-cost, power-efficient Object Following Robot can be successfully built using the Arduino UNO with combined acoustic and infrared sensing, without requiring expensive vision-based hardware.

Topic 7: Troubleshooting & Future Scope

Q129. What problem was observed with the servo motor in the initial prototype?

Answer: The SG90 servo motor drew a high inrush current during rotation, causing a voltage brownout on the 5V logic rail that repeatedly reset the Arduino and produced erratic sensor data.

Q130. What component is proposed to fix the servo motor voltage problem in future versions?

Answer: An LM2596 step-down buck converter would provide a separate, isolated power rail for the servo motor, preventing its transient current draw from reaching the Arduino's 5V logic rail.

Q131. What is the difference between a linear regulator (LM7805) and a buck converter (LM2596)?

Answer: A linear regulator dissipates excess voltage as heat (low efficiency), while a buck converter uses high-frequency switching and an inductor to transfer energy efficiently (much higher efficiency).

Q132. What is the proposed upgrade to add a wider sensor scanning field of view?

Answer: Re-integrating the SG90 micro servo to rotate the HC-SR04 ultrasonic sensor through up to 180 degrees, creating an active scanning arc instead of a fixed forward-looking beam.

Q133. What would re-introducing the OLED display add to the system?

Answer: A 0.96-inch SSD1306 OLED display would show the current target distance and robot action (Forward, Backup, Searching) in real time, allowing users to monitor robot behavior without a PC.

Q134. What microcontroller upgrade is proposed for wireless and camera-based tracking?

Answer: Upgrading to an ESP32 with an ESP32-CAM module would enable Wi-Fi, Bluetooth, and onboard camera-based human detection, eliminating the need for dedicated IR and ultrasonic hardware.

Q135. What is a PID controller and why is it proposed as a future improvement?

Answer: A Proportional-Integral-Derivative (PID) controller continuously adjusts motor speed based on the error between the desired and actual position, producing smoother acceleration and deceleration.

Q136. What is the major limitation of the L298N motor driver in terms of power efficiency?

Answer: The L298N uses BJT transistors with a high saturation voltage drop of approximately 1.8V to 2.0V, wasting battery energy as heat and reducing effective motor torque.

Q137. What modern component would replace the L298N for better efficiency?

Answer: A MOSFET-based motor driver such as the TB6612FNG, which has a much lower ON-resistance ($R_{DS(on)}$) and significantly less heat dissipation than the BJT-based L298N.

Q138. What is the difference between a Time-of-Flight (ToF) sensor and the HC-SR04?

Answer: A ToF sensor uses laser light pulses and measures the return time of photons, providing faster and higher-precision distance measurements unaffected by surface material or temperature.

Q139. What is sensor fusion, and how could it improve this robot?

Answer: Sensor fusion combines data from multiple sensors (e.g., ultrasonic, camera, ToF) using algorithms to produce a more accurate and reliable model of the environment than any single sensor alone.

Q140. What would happen if the common ground connection between the Arduino and L298N was removed?

Answer: The digital control signals (IN1–IN4, ENA, ENB) from the Arduino would have no shared voltage reference with the L298N driver, causing all motor direction logic to fail or behave erratically.